

Donghu Kim

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Research Interest

Efficient Reinforcement Learning

- Making RL work with as little samples and/or compute as possible.
- Related: [Simba](#), [SimbaV2](#), [FlashSAC](#)

Complex Dexterous Manipulation

- Achieving dexterous manipulation tasks via RL in simulation.
- Related: [FlashSAC](#)

Education

KAIST

M.S. in Artificial Intelligence (GPA: 3.82/4.3, Advisor: Jaegul Choo)

Seongnam, Korea
Mar. 2024 - Present

Korea University

B.S. in Computer Science (Major GPA: 4.5/4.5)

Seoul, Korea
Mar. 2018 - Feb. 2024

Work Experience

Holiday Robotics

AI Research Scientist

Seoul, Korea
Jan. 2026 - Present

- Dexterous manipulation via reinforcement learning.

Krafton AI

Research Intern @ Physical Intelligence Team

Seoul, Korea
Jun. 2025 - Sep. 2025

- Developed locomotion RL policy for Unitree G1 robots to navigate company environment.

Publications & Preprints

FlashSAC: Fast and Stable Off-Policy RL for High-Dimensional Robot Control

ArXiv'26

Donghu Kim*, Youngdo Lee*, Minho Park, Kinam Kim, I Made Aswin Nahendra, Takuma Seno, Sehee Min, Daniel Palenicek, Florian Vogt, Danica Kragic, Jan Peters, Jaegul Choo, Hojoon Lee

Preprint

- [arXiv](#) / [project page](#) / [github](#) (★100+)

FIRE: Frobenius-Isometry Reinitialization for Balancing Stability-Plasticity Tradeoff

ICLR'26

Isaac Han, Sangyeon Park, Seungwon Oh, Donghu Kim, Hojoon Lee, Kyungjoon Kim

Oral

- [arXiv](#) / [project page](#) / [github](#)

Building Resource-Constrained Language Agents

EMNLP'25

Hojun Cho, Donghu Kim, Soyoung Yang, Chan Lee, Hunjoo Lee, Jaegul Choo

Poster

- [arXiv](#)

SimbaV2: Hyperspherical Normalization for Scalable Deep Reinforcement Learning

ICML'25

Hojoon Lee*, Youngdo Lee*, Takuma Seno, Donghu Kim, Peter Stone, Jaegul Choo

Spotlight

- [arXiv](#) / [project page](#) / [github](#)

Simba: Simplicity Bias for Scaling Up Parameters in Deep Reinforcement Learning

ICLR'25

Hojoon Lee*, Dongyoon Hwang*, Donghu Kim, Hyunseung Kim, Jun Jet Tai, Kaushik Subramanian, Peter R. Wurman, Jaegul Choo, Peter Stone, Takuma Seno

Spotlight

- [arXiv](#) / [project page](#) / [github](#)

Dynamic Mixture of Experts Against Severe Distribution Shifts

Donghu Kim

- [arXiv](#)

ArXiv'25

Preprint

ATARI-PB: Investigating Pre-Training Objectives for Generalization in Pixel-Based RL

Donghu Kim*, Hojoon Lee*, Kyungmin Lee*, Dongyoon Hwang, Jaegul Choo

- [arXiv](#) / [project page](#) / [github](#)

ICML'24

Poster

Slow and Steady Wins the Race: Maintaining Plasticity with Hare and Tortoise Networks

Hojoon Lee, Hyeonseo Cho, Hyunseung Kim, **Donghu Kim**, Dugki Min, Jaegul Choo, Clare Lyle

- [arXiv](#) / [github](#)

ICML'24

Poster

Projects

KAN RL

2024

- Implemented Kolmogorov-Arnold Network in sequential Atari environments and investigated its relevance to catastrophic forgetting and plasticity.
- [report](#) (Colab)

RL Basic Tutorial

2024

- Developed and delivered a series of three lectures on reinforcement learning for a government-funded bootcamp program in Korea.
- [program page](#) / [material1](#) (Korean) / [material2](#) (Korean)

Character-level BERT

2022

- Proposed a character-level tokenizer for BERT to enhance robustness against character-level attacks common in spam emails.
- [report](#) / [code](#)

Honors & Awards

Dean's List (4.5/4.5 semester GPA)

Korea University

2019, 2022

Starcraft AI Competition Silver Prize (\$2000)

NCsoft AI Fellowship

2019

Presidential Science Scholarship (\$40000 in total)

Korea Student Aid Foundation

2018

Academic Services

Reviewer

International Conference on Learning Representations (**ICLR**)

2025 - 2026

Reviewer

Conference on Neural Information Processing Systems (**NeurIPS**)

2025 - 2026